

Selma Mazioud

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Education

UC Berkeley, <i>PhD in Statistics</i>	Expected 2030
Yale University, <i>BS in Applied Mathematics, with distinction</i>	May 2025

Publications, presentations, and preprints

Deriving Causal Subspaces in Transformers Andrew Jun Lee, Selma Mazioud, Kyle J Ray, Jasmina Urdshals, Paul M. Riechers, Adam Shai <i>Interpretability as a Science Workshop at NeurIPS 2026</i>	December 2026
Tracking Training Phases in Compositional Learning with Task-Agnostic Measures Niclas Dern*, Selma Mazioud* , Jakob Heiss, Avrajit Ghosh, Curtis James McDonald, Gabriel Clara, Bin Yu. <i>Mechanistic Interpretability Workshop at ICML 2026</i>	August 2026
Exploring the Manifold of Neural Networks using Diffusion Geometry Elliott Abel*, Drew Steindl*, Selma Mazioud* , Ellie Schueler*, Folu Ogundipe*, Ellen Zhang, Yvan Grinspan, Kristof Reimann, Peyton Crevasse, Dhananjay Bhaskar, Kaly Zhang, Tim Rudner, Ian Adelstein, Smita Krishnaswamy. <i>arXiv preprint, arXiv</i>	October 2024
Sequentially Charming Chemical Freeze-out in Relativistic Heavy Ion Collisions at the LHC and RHIC Fernando Flor, Selma Mazioud . 2024 Fall Meeting of the American Physical Society, Division of Nuclear Physics. Abstract	October 2024
Statistical Hadronization Model Calculations of Heavy Flavor Hadron Production in Relativistic Heavy-Ion Collisions at RHIC and the LHC Selma Mazioud , Fernando Flor. XXXth International Conference on Ultra-relativistic Nucleus-Nucleus Collisions. Abstract and poster	September 2023

Research experience and projects

Machine Alignment, Transparency, and Security (MATS). ◦ MATS 10.0 Scholar. Advised by Paul Riechers and Adam Shai (Simplex).	June—August 2026
UC Berkeley. Yu Group. ◦ Graduate Student Researcher. <i>Tracking Phase Transitions in Compositional Learning with Task-Agnostic Measures.</i> Advised by Berkeley Professor Bin Yu. – 6-month project funded by the UK AI Security Institute Alignment Project grant.	Jan—May 2026
Wu Tsai Institute. Krishnaswamy Laboratory. ◦ Senior project. <i>Measuring Entropy in Graph Datasets Using Geometric Scattering and Diffusion Geometry.</i> Advised by Yale Professors Smita Krishnaswamy and Ian Adelstein. Thesis – Proposed a novel method to compute entropy over a set of graphs, providing a statistic that quantifies the variability of a graph dataset – Demonstrated that our method effectively captures the variance of a graph dataset through experiments on toy datasets generated by increasingly perturbing a graph.	June 2024—May 2025

- **Yale REU.** *Exploring the Manifold of Neural Networks Using Diffusion Geometry*. Advised by Yale Professors Smita Krishnaswamy and Ian Adelman.
 - Showed that high-performing neural networks, across architectures, exhibit higher diffusion spectral entropy in their hidden embeddings of the data.
 - Using the eigenspectrum of the manifold graph Laplacian, showed that test accuracy and learning rate are low-frequency signals, which indicates we are able to interpolate in the space of neural networks to choose parameter ranges.

Wright Laboratory. Relativistic Heavy-Ion Group (Caines Group).

May 2023—May 2025

- **Class project.** *Graph-based Machine-Learning Methods for Particle Reconstruction at the LHC*. Advised by Yale Professor Helen Caines. [Code and report](#) 
- Utilize a trainable adjacency matrix to embed node features into a highly expressive latent space where clustering is easier to deploy.
- **Research project for credit.** *Machine-Learning Methods for Particle Reconstruction at the LHC*. Advised by Yale Professor Helen Caines.
 - Demonstrated that an integrated approach combining quadratic regression and k-means clustering not only attained high accuracy but also exhibited remarkable robustness against Gaussian noise.
- **Yale REU.** *Statistical Hadronization Model Calculations of Charm Hadron Production in Relativistic Heavy Ion Collisions at the LHC*. Advised by Yale Professor Helen Caines.
 - Modeled the relationship between fugacity factor, which accounts for the production of charmed hadrons resulting from the initial collision scattering, and temperature of chemical freezeout.
 - Showed that charm quarks hadronize at higher temperature than strange and light quarks.

Fellowships and awards

Berkeley Fellowship for Graduate Study; 2025-2027

Morse Head of College Leadership Award; 2025

Yale Science and Engineering Association Award for Outstanding Undergraduate Achievement; 2025

McCall MacBain Scholarship Finalist; 2025

Summer Undergraduate Research in Mathematics at Yale (SUMRY) fellowship; 2024

Yale Richter fellowship (Summer Undergraduate Research Fellowship); 2023, 2024

Teaching Experience

Undergraduate Learning Assistant for Theory of Statistics	Spring 2025
Course-based Peer Tutor for Calculus II	Fall 2023—Spring 2025
Undergraduate Learning Assistant for Introductory Machine Learning	Fall 2024
Undergraduate Learning Assistant for Algorithms	Spring 2024
Undergraduate Learning Assistant for Data Analysis and Exploration	Spring 2023, Fall 2023
French Language Tutor	Fall 2022, Spring 2023

Service and Leadership

Applied Mathematics Departmental Student Advisory Committee	Fall 2024, Spring 2025
Middle Eastern and North African Student Association	August 2022—May 2025
◦ Secured a \$60,000 budget for Yale's first full MENA space (the first of its kind in the Ivy League) and hired a full-time MENA Assistant Director. Yale President Announcement . The Yale Daily News . The Yale Daily News . The Yale Herald . 	